



Salvatore T. Ciminello

(914) 268-7025 | stc92@cornell.edu | [linkedin.com/in/sal-ciminello](https://www.linkedin.com/in/sal-ciminello) | [Portfolio](#) ↗

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineering, GPA: 3.71
Master of Engineering, Aerospace Engineering, GPA: 4.02

May 2026

Expected December 2026

Relevant Courses (* = M.Eng Graduate Level):

Dynamics and Controls: Intermediate Dynamics* | Mechatronics | System Dynamics | Autonomous Mobile Robotics*
Spacecraft Architecture: Propulsion of Aircraft & Rockets | Space Structures* | Spacecraft Systems Architecture*

Teaching Experience: System Dynamics (Lab Asst.) | Mechatronics (TA) | Dynamics (Grader)

TECHNICAL SKILLS

Hardware: Machining (Mill and Lathe), Rapid Prototyping, Additive Manufacturing, Electronics and Sensor Wiring, Feedback Control Systems

Software: MATLAB, Python, C++, Arduino, Raspberry Pi

CAD / Simulation: SolidWorks, Fusion 360, ANSYS Mechanical, ANSYS Maxwell, ANSYS Fluent

TECHNICAL EXPERIENCE

Cornell Aerospace Adversary Lab – Research Assistant, M.Eng Technical Project

January 2026-Present

Air Force Research Laboratory (AFRL) funded | Prof. Falco (cybersecurity in space)

- Designing and building physical CubeSat structural mockups for a distributed constellation testbed; validating structural layout and intersatellite communication links under variable formation configurations.
- Implementing distributed formation control to coordinate multi-satellite positioning, supporting hardware-in-the-loop testing of network-level attack propagation across the constellation.

Active and Passive Pod Control System – Hyperloop Project Team

August 2024-March 2026

- Derived passive roll stability system by modeling pod as a spring-damper about its neutral axis via momentum balance; selected shock absorber and caster wheel configuration by calculating required damping coefficients analytically.
- Developed 1–4 DOF MATLAB simulations of active electromagnetic levitation, targeting a 4–5 mm air gap; characterized electromagnet force-distance behavior through physical testing and fit an exponential model to experimental data.

Autonomous Mobile Robot – Autonomous Mobile Robotics, Cornell University

January 2026 - May 2026

- Implemented robot localization using Kalman, Extended Kalman, and Particle Filters; developed RRT-based path planning for competitive autonomous navigation challenges.

Robotic Arm (in progress)

June 2026 - Present

- Designing and building a 6 DOF robotic arm end-to-end: mechanical design in Fusion 360, Raspberry Pi 5 + PCA9685 servo control, inverse kinematics in Python; currently finishing mechanical assembly.
- Designed wrist roll joint using servo's center thread for axial load retention; upgraded proximal joint to a 25 kg·cm servo after torque calculations showed the stock servo undersized for the load.

Autonomous Rescue Drone (in assembly)

June 2026 - Present

- Custom 3D-printed shell over an F450 frame, Holybro Pixhawk 6C Mini running ArduPilot; full mechanical build and electrical integration self-performed, including soldering XT60 power connections, and RadioMaster Pocket ELRS receiver.

LEADERSHIP EXPERIENCE

Mechanical Team Lead – Cornell Hyperloop Project Team

May 2025-Present

- Led ~40 engineers across six subteams, including Structures, Magnetics, Braking, Composites, Rapid Prototyping, and Electromagnetic Levitation; coordinated closely with a 20-person Electrical team and attended weekly cross-functional leads meetings spanning mechanical, electrical, and business.

Magnetics & Propulsion Subteam Lead – Cornell Hyperloop Project Team

December 2024-May 2025

- Directed levitation, controls, propulsion, and cooling projects; designed multi-axis PID levitation control system using 4 proximity sensors and PWM controllers to regulate height and attitude across all 3 axes.